1 A student is investigating the reflection of light from a compact disc (CD), as shown in Fig. 1.1.

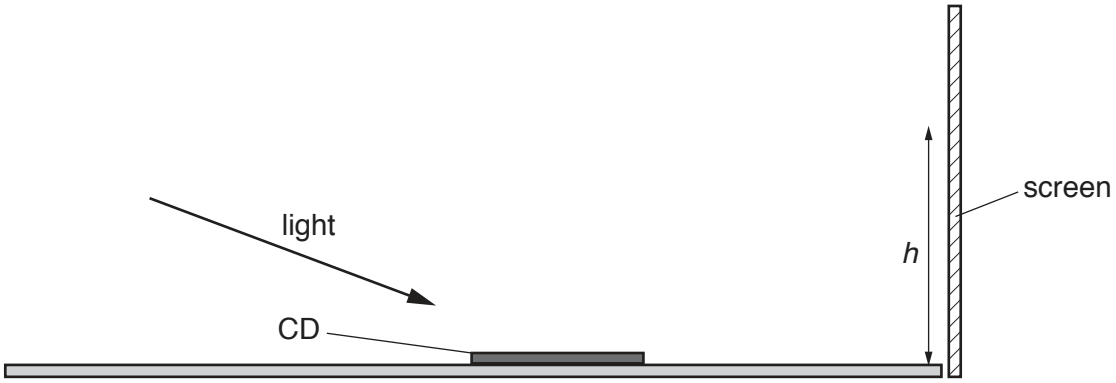


Fig. 1.1

The student observes on the screen a pattern of maxima and minima which is similar to that produced by a diffraction grating. The distance h is measured to one of the maxima.

It is suggested that the relationship between h and the wavelength λ of the incident light is

$$h = \frac{n\lambda}{d} + B$$

where n is the order of the maximum and d and B are constants.

Design a laboratory experiment to test the relationship between h and λ.
Explain how your results could be used to determine values for d and B.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to

- the procedure to be followed,
- the measurements to be taken,
- the control of variables,
- the analysis of the data,
- any safety precautions to be taken.

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